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INF 651 Front End I

12/12/2024

Interactive Expense Tracker and Budgeting Tool

Introduction

This document describes the development of an interactive web application for tracking expenses and managing budgets. The application was created using **HTML, CSS, and JavaScript** to provide dynamic and responsive functionality. It allows users to add, view, and manage their expenses with real-time updates and validations.

Features and Functionality

1. Home Page (Dashboard)

- **Summary Display:** Displays a summary of total income, total expenses, and the remaining budget.
- **Dynamic Updates:** The dashboard updates dynamically whenever new expenses are added.
- **Navigation Bar:** Links to other sections of the application (Add Expense, View Expenses and Home).

2. Add Expense Page

- **Form Input:** Users can enter the category, amount, and description of an expense.
- **Real-Time Validation:** Ensures valid amounts and non-empty descriptions are entered.
- **Dynamic Submission:** Upon submission, the expense is added to the list, and the dashboard updates instantly.

3. View Expenses Page

- **Expense List:** Displays all recorded expenses with details like category, amount, and description.

- **Dynamic Rendering:** Updates the list dynamically as expenses are added.
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How JavaScript Enhances the User Experience

1. DOM Manipulation:

- JavaScript dynamically updates the dashboard and expense list without reloading the page.
- When a user adds or deletes an expense, the changes are instantly reflected in the interface.

2. Event Handling:

- Click events are used for submitting the expense form.
- The application responds immediately to user actions, such as adding or viewing expenses and set the total income.

3. Input Validation:

- Ensures users enter valid data (e.g., positive numbers for expense amounts).
- Provides alerts if incorrect data is entered, enhancing data integrity.

4. Data Storage and Retrieval:

- Uses the browser's **local Storage** to store expenses.
- Allows the application to retain data even after the page is refreshed.

5. Reusable Functions:

- Functions for adding, displaying, and updating expenses make the code modular and maintainable.
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Challenges Faced and How They Were Overcome

1. **Issue:** Ensuring data persistence across different pages.

- **Solution:** Implemented **localStorage** to save expense data, ensuring it persists even after a page reload.

2. **Issue:** Real-time updates to the dashboard when new expenses are added.

- **Solution:** Created functions to update the dashboard dynamically after each expense submission.
 - 3. **Issue:** Validating user inputs in the expense form.
 - **Solution:** Added JavaScript checks to ensure amounts are positive numbers and descriptions are not empty.
 - 4. **Issue:** Ensuring user inputs total income from the home page
 - **Solution:** Added a button in the home page for the user to be able to update the total income anytime.
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Future Plans for Additional Features or Backend Integration

1. **Backend Integration:**
 - Implement a **database (MySQL)** to store expense data permanently.
 - Enable user authentication to allow personalized expense tracking.
 2. **Visual Analytics:**
 - Add **charts and graphs** to visualize spending patterns.
 - Use libraries like **Chart.js** to create dynamic visualizations.
 3. **Savings Goals:**
 - Implement goal-setting features with progress tracking and notifications.
 - Alert users when they are close to meeting or exceeding their savings goals.
 4. **Currency Converter:**
 - Integrate a **currency conversion API** to support international users.
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Conclusion

The Expense Tracker web application demonstrates effective use of JavaScript for creating dynamic and interactive features. It provides users with a functional and engaging tool for managing their finances. With future enhancements, this application can become an even more powerful budgeting solution.

